

Figure 2-3: Negative (falling) – Positive (rising) edge triggering

Assignment 3

The aim of this part of the lab was to learn how to use the trigger knob on the oscilloscope. The knob is designed to use for stabilizing the waveform on the oscilloscope screen. Following the instructions, a 1 Vpp triangular wave with 2 kHz frequency were applied. Observations are shown below.



Figure 4-5: Waveform before-after being stabilized

Assignment 4

A digital to analog converter (DAC) converts digital information into analog information. An analogue to digital converter (ADC) is a device that converts analog data into digital data. The input to the signal generator is displayed digitally on the oscilloscope in this experiment. The analogue data supplied by the signal generator is the input, therefore it's easy to see how ADC is employed in the oscilloscope that was used in this experiment. The experiment's next step was to try out all of the different acquisition modes (sample, peak detect, average). The oscilloscope is first given a 1 Vpp square wave at a frequency of 5 kHz. "The acquisition modes of an oscilloscope control how waveform points are generated from sample points, which refer to the digital values taken directly from the analogue to digital converter (ADC). In peak detect mode, the oscilloscope utilizes the maximum and minimum sample points saved during two waveform intervals, as its corresponding waveform points. Sample mode creates a waveform point in the oscilloscope by saving one sample point during every waveform interval. The oscilloscope, in average mode, saves a sample point taken during a waveform interval. The final displayed waveform is produced by averaging the waveform points from consecutive acquisitions."¹

¹ <https://www.tek.com/document/online/primer/xyzs-scopes/ch4/oscilloscope-systems-and-controls>

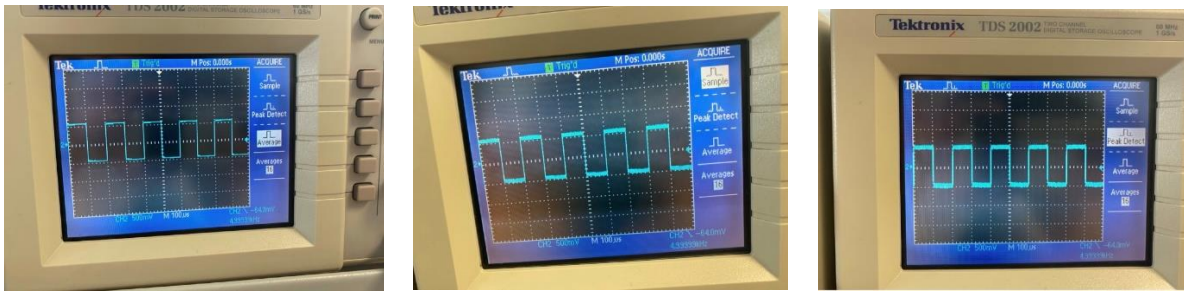


Figure 6-7-8: acquisition modes average, sample, peak detect

Assignment 5

The purpose of this experiment is to understand what DC offset, AC coupling and DC coupling are. A sinusoidal signal with 2 Vpp amplitude and 1 kHz frequency were applied. AC (alternating coupling) allows only AC signals to pass through a connection, but DC (direct coupling) allows both AC and DC signals from 0 Hz up to the bandwidth of the scope pass through.² Because 1 V of DC offset was applied at the beginning of the experiment, there is a 1 V discrepancy between the peaks.

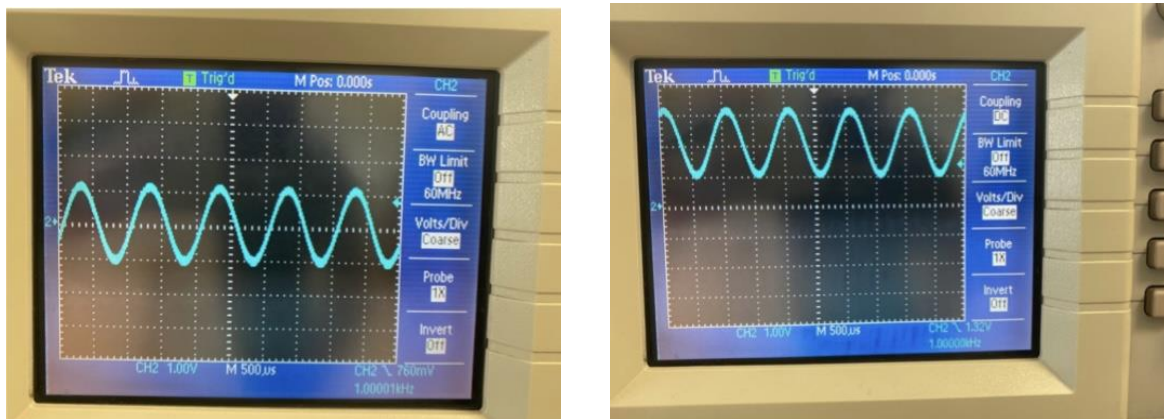


Figure 9-10: AC-DC Coupling modes

² <https://community.sw.siemens.com/s/article/ac-and-dc-coupling-what-s-the-difference>

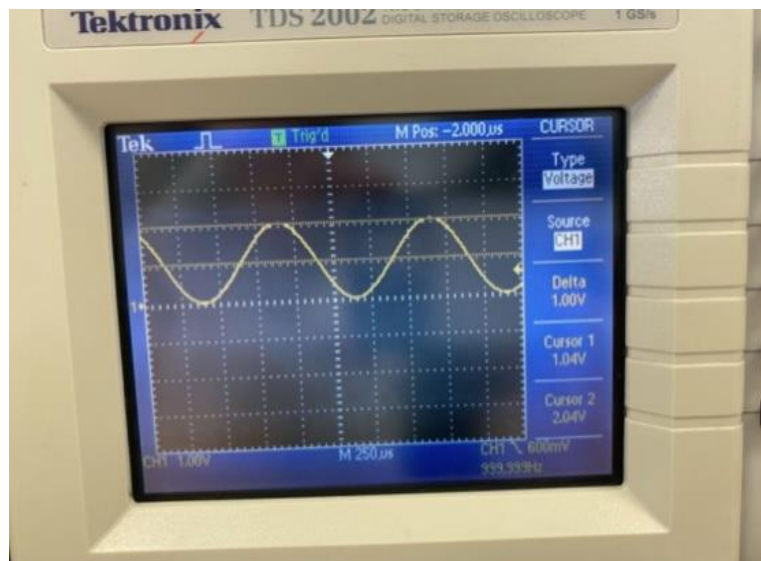


Figure 11: Voltage difference between AC-DC Couplings

Assignment 6

Breadboards are used to connect electronic components with through-hole ports. Long metal leads are meant to be inserted through holes in a printed circuit board (PCB) covered with a thin copper layer, allowing the components' leads to be soldered to the board.³

As the X signal, a 2 Vpp 1 kHz sinusoidal signal (with 0 DC offset) is used to start the experiment. The oscilloscope displays both X and Y signals using a second probe. Both channels' grounds were connected to a common ground. Channel 1 is for X, while channel 2 is for Y. The source of the trigger is Channel 1. The time difference between signals X and Y was 300 μ s. The frequency was modified to 100 kHz after detecting the delay. It's been observed that there appeared a visible difference in the delay on the waveform where actually there shouldn't be. The draw was changed which could have been occurred due to the error of the oscilloscope measurements.

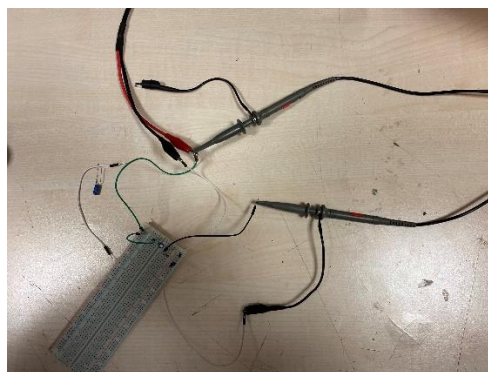


Figure 12: Circuit on the breadboard

³ <https://www.sciencebuddies.org/science-fair-projects/references/how-to-use-a-breadboard>

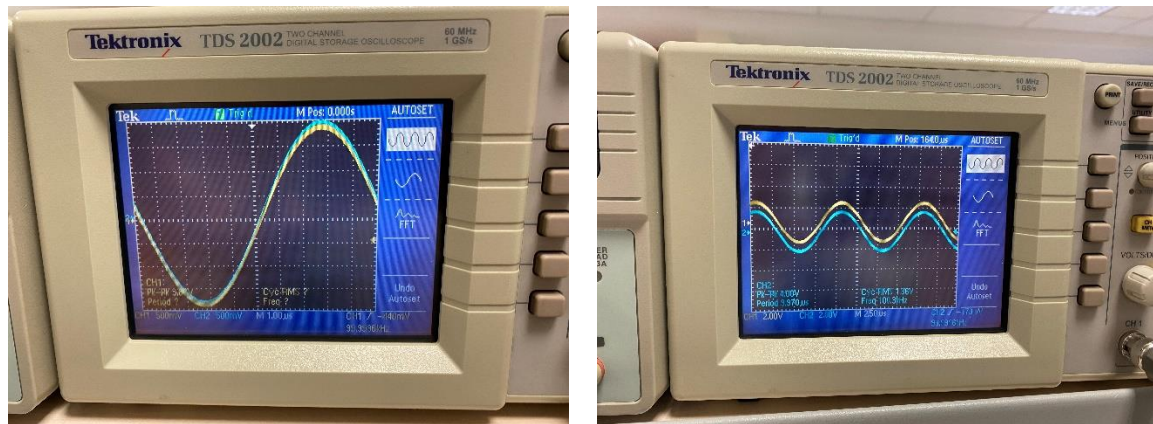


Figure 13-14: Waveforms when frequency at 1KHz and 100KHz